

Interacting Multipoles of Rank 1 and 3:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \delta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \delta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \gamma) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \delta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \gamma) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \beta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \end{array} \right] \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Omega_{\beta\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Omega_{\beta\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{array} \right] \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha \cdot \Omega_{\beta\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\delta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{array} \right] \quad (2b)$$

Simplify R:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{zzzz} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{z\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{z\beta\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{z\beta\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{array} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} -15 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{array} \right] \quad (3b)$$

Simplify Multipoles:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\beta\beta} \end{array} \right] \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{l} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\beta\beta} \end{array} \right] \quad (4b)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \quad (5a)$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \quad (5b)$$

Reduce Indices:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \quad (6a)$$

$$\xRightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \quad (6b)$$

Add up:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot \frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \quad (7)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = \begin{bmatrix} -3 \cdot R_z^{-5} \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \quad (8)$$